People's Education Trust(R.),

P.E.S. College of Engineering, Mandya

(An Autonomous Institution Aided by Government of Karnataka,Affiliated to VTU,Belagavi)

NEXOVATE WATER ROCKET

OBJECTIVES:

Step into the world of aerospace innovation with our exhilarating Water Rocket event! Water Rocket Event is an exciting technical competition where participants design and launch rockets using water propulsion.The primary goal is to engineer a rocket capable of achieving the highest possible performance score based on flight physics. This event is specifically designed to test a team's creativity, their understanding of aerodynamics, and their ability to design for flight stability in a competitive environment.

ELIGIBILITY:

•The competition is open to all students.
•Each team must consist of 2 to 4 members.
•A participant cannot be a part of multiple teams.

Open to all students from any college or institution. Participants can compete individually or as a team, as per event rules. All participants ust carry a valid student ID card for verification. Entry will be permitted only after successful registration. Participants are expected to follow all event guidelines and maintain discipline throughout the competition.

PESCE MANDYA THE ROBO HUB

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INSTITUTION'S INNOVATION COUNCIL

NEXOVATE

GAMEPLAY & ADVANCED SCORING :

1.General Rules of Engagement :

- Team Size: Each team may consist of a maximum of 3 members.
- Inventory: Only one rocket is allowed per team throughout the event.
- Punctuality: Teams must report to the launch site at least 15 minutes early for technical inspection.

THE DUAL-CHALLENGE FORMAT

Challenge I: Maximum Distance Sprint

This event tests the maximum thrust and aerodynamic efficiency of the design.

- The Launch: Rockets are pressurized to a limit set by the officials. Teams may choose their launch angle.
- Scoring Goal: Achieve the longest horizontal displacement within the designated lane and with maximum time.

Total Points = Distance (meters) X Time of Flight (seconds)

- Distance: Measured from the launch pad to the point of first impact.
- Time of Flight: Measured from the moment of lift-off until the rocket touches the ground.
- Metric recording: Performance data is measured and recorded immediately after each of the two attempts.

Challenge II: Archery-Style Target Precision

This event tests the team's ability to throttle their rocket for accuracy. Participants must adjust water volume and pressure to land in a specific zone.

The Target: A series of concentric rings (Archery Style) is marked on the field at a set distance.

Point Specifications per Ring:

- Center: Highest point value.
- Inner Ring: Mid-high point value.
- Outer Ring: Base point value.

Measurement: The score is determined by the final resting position of the rocket's nose cone within the rings. If the rocket lands outside all rings, it receives zero points for this round

NEXOVATE

ROCKET CONSTRUCTION SPECIFICATIONS

To ensure fair competition, all entries must adhere to the following build requirements: — —

- Pressure Vessel: The main body of the rocket must be constructed using PET bottles. —
- Propulsion: The rocket must be powered strictly by a combination of water and pressurized air. —
- Material Safety: The use of any harmful or dangerous materials in the construction is strictly prohibited. —
- Structural Integrity: Every rocket must possess a stable structure to ensure a safe and predictable flight path.

JUDGING & DISQUALIFICATION

Evaluation & Winner Selection

Judicial Authority:

The decisions made by the judges regarding scoring and rule compliance are final. The winners are determined by the combined performance in both challenges.

Tie-Breaker:

In the event of a total score tie, the team that scored higher in the Archery Target Challenge will be ranked higher, as it demonstrates superior control and calculation